

Mahnoor Zaffar

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BACKEND ENGINEER | AI/ML
ENGINEER

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EDUCATION

National University of Sciences and Technology

September 2026

MS. Computational Sciences and Engineering

[Access Portfolio](#)

PROFESSIONAL WORK EXPERIENCE

Full Stack Engineer

Nov 2025 — Jan 2026

High Mark International (US Client | NDA)

- Built and maintained production web applications using React and Python, developing REST APIs and Redis-backed services while supporting automation workflows processing 10,000+ job listings.

Freelance Backend / Full Stack Engineer

June 2023 — Present

We Work Remotely

- Delivered client applications and backend systems using Python, FastAPI, Next.js, PostgreSQL, Docker, and OpenAI APIs across short-term engineering projects. Working with Australian Clients on projects.

Machine Learning Engineering Intern

March 2026 — July 2026

The Buraq Association

- Developed and evaluated machine learning solutions using Python, building data preprocessing, model training, experimentation, and evaluation workflows.

TECHNICAL SKILLS

Languages

Python, JavaScript, TypeScript, SQL

Backend

FastAPI, REST APIs, SQLAlchemy, JWT, Redis, Celery, PostgreSQL, Supabase, NoSQL, WebSockets

Frontend

Next.js, React, Tailwind CSS, HTML, CSS

AI/ML

OpenAI APIs, LLMs, RAG, Prompt Engineering, AI Agents, Tool Calling, MCP, Vector Databases, Embeddings, Semantic Search, LLM Evaluation

DevOps

Docker, Docker Compose, Git, GitHub Actions, Linux

Tools

GitHub, Playwright, OpenRouter

PROJECTS

ContainerOps AI | Python • FastAPI • Docker • MCP • LLM Agents

- Built an agentic infrastructure recovery system that diagnoses and remediates failures in containerized applications by inspecting logs, health status, configuration, and runtime state through scoped tools.
- Implemented an **Observe** → **Diagnose** → **Act** → **Verify** workflow with explicit tool permissions, recovery verification, and action auditing to constrain the blast radius of incorrect LLM decisions.

Tayari.ai | FastAPI • Next.js • PostgreSQL • Redis • Supabase • OpenAI

- Built an AI-powered technical interview platform with a modular FastAPI backend for authentication, interview orchestration, conversation management, and structured LLM evaluation.
- Implemented streaming interview sessions, rubric-based evaluation, conversation summarization, and Redis caching to reduce context overhead and improve response latency.
- Designed asynchronous evaluation workflows with PostgreSQL persistence to support concurrent interview sessions and reliable result processing.

VocalFlux | Python • FastAPI • WebSockets • Whisper • Next.js • Docker

- Built a real-time speech-to-text system using FastAPI WebSockets and Whisper, streaming PCM audio from the browser through an inference pipeline and returning incremental transcriptions.
- Implemented audio chunking, concurrent session handling, and Dockerized CPU/GPU inference to support reproducible real-time speech processing.

Job Hunting Platform | FastAPI • Celery • Playwright • PostgreSQL • OpenRouter

- Built an asynchronous job aggregation platform integrating multiple job boards and ATS sources, processing **1,000+ job listings daily** through concurrent Playwright workers.
- Implemented PostgreSQL-backed data normalization and persistence to unify listings across sources and prevent duplicate records.

CERTIFICATIONS

- Professional Certificate in AI Engineering DataCamp
- Full Stack Engineer Codecademy
- AI/ML Engineer Codecademy

March 2026
July 2026
August 2026

ACHIEVEMENTS

- Built 40+ full-stack applications for diverse clients using Next.js, FastAPI, and PostgreSQL.
- Reduced deployment time by 60 percent by implementing CI/CD pipelines with GitHub Actions.
- Developed internal business tools automating manual processes, saving 20+ hours/week.
- Developed production-ready AI workflow automation systems.

OPEN SOURCE CONTRIBUTIONS

- **Hugging Face PEFT** – Fixed base-weight corruption when deleting merged LoRA/adapters by adding an auto-unmerge guard to `BaseTunerLayer.delete_adapter()`, with regression tests across adapter types. *Open, under review.*
- **vLLM** – Diagnosed and fixed an NVFP4 checkpoint shape-mismatch bug in `flashinfer_scaled_fp4_mm`. *Open, under review.*
- **LlamaIndex** – Extracted JSON Schema-to-Pydantic conversion into a standalone, client-independent module to improve modularity and reuse. *Open, under review.*
- **sktime** – Fixed a CPU deserialization bug in `MantisClassifier` by adding `map_location` to `torch.load()` calls. *Open, under review.*
- **pandas** – Added an RFC 9116 `security.txt` file to improve vulnerability-disclosure discoverability. *Open, under review.*